

HW1

$$1. \hat{y}_{knn} = \frac{y_1 + y_2 + y_3}{3}$$

estimate 150:

150	→	65 kg
168	→	78 kg
171	→	80 kg

$$\hat{y}_{150} = \frac{65 + 78 + 80}{3} \approx 74.33 \text{ kg}$$

estimate 155:

150	→	65 kg
168	→	78 kg
171	→	80 kg

$$\hat{y}_{155} = \frac{65 + 78 + 80}{3} \approx 74.33 \text{ kg}$$

estimate 165:

168	→	78 kg
171	→	80 kg
178	→	82 kg

$$\hat{y}_{165} = \frac{78 + 80 + 82}{3} \approx 80.33 \text{ kg}$$

estimate 180:

178	→	83 kg
182	→	80 kg
191	→	100 kg

$$\hat{y}_{180} = \frac{83 + 80 + 100}{3} \approx 87.67 \text{ kg}$$

$$2. \hat{y}_{knn} = \frac{w_1 y_1 + w_2 y_2 + w_3 y_3}{w_1 + w_2 + w_3}$$

$$w_i = \frac{1}{d_i}$$

150:

$ 150 - 150 = 0$	$w_1 = \frac{1}{0}$	$\hat{y}_{150} = 65 \text{ kg}$
$ 150 - 168 = 18$	$w_2 = \frac{1}{18}$	
$ 150 - 171 = 21$	$w_3 = \frac{1}{21}$	

155:

$ 155 - 150 = 5$	$w_1 = \frac{1}{5}$
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$ 155 - 168 = 13$	$w_2 = \frac{1}{13}$
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$ 155 - 171 = 16$	$w_3 = \frac{1}{16}$
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$$\hat{y}_{155} =$$

$$\frac{\frac{1}{5} \times 65 + \frac{1}{13} \times 78 + \frac{1}{16} \times 80}{\frac{1}{5} + \frac{1}{13} + \frac{1}{16}} \approx 70.71 \text{ kg}$$

$$165: |165 - 168| = 3 \quad w_1 = 1/3$$

$$|165 - 171| = 6 \quad w_2 = 1/6$$

$$|165 - 178| = 13 \quad w_3 = 1/13$$

$$\hat{y}_{165} = \frac{\frac{1}{3} \times 78 + \frac{1}{6} \times 80 + \frac{1}{13} \times 83}{\frac{1}{3} + \frac{1}{6} + \frac{1}{13}} \approx 79.24 \text{ kg}$$

$$190: |190 - 178| = 12 \quad w_1 = 1/12$$

$$|190 - 182| = 8 \quad w_2 = 1/8$$

$$|190 - 191| = 1 \quad w_3 = 1$$

$$\hat{y}_{190} = \frac{\frac{1}{12} \times 83 + \frac{1}{8} \times 80 + 1 \times 100}{\frac{1}{12} + \frac{1}{8} + 1} \approx 96.76 \text{ kg}$$

3. $J(x) = x^T Q x + d^T x + c \quad Q = Q^T$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & c \\ c & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} ax_1 + cx_2 \\ cx_1 + bx_2 \end{bmatrix}$$

$$\nabla_x J(x): \quad x^T Q x = \sum_{i=1}^n \sum_{j=1}^n q_{ij} x_i x_j$$

$$x_1(ax_1 + cx_2) + x_2(cx_1 + bx_2) \\ ax_1^2 + cx_1x_2 + cx_1x_2 + bx_2^2$$

$$\frac{\partial (x^T Q x)}{\partial x_k} = \sum_{i=1}^n q_{ik} x_i + \sum_{j=1}^n q_{kj} x_j = Q x_k + Q^T x_k = 2Q x_k$$

$$\frac{\partial (d^T x)}{\partial x} = d \quad \frac{\partial c}{\partial x} = 0$$

$$\Rightarrow \nabla_x J(x) = 2Qx + d$$

show Hessian: $\frac{\partial (2Qx + d)}{\partial x} = 2Q$

$$2Qx + d = \begin{bmatrix} 2Q_{11}x_1 + 2Q_{12}x_2 + d \\ 2Q_{21}x_1 + 2Q_{22}x_2 + d \end{bmatrix}$$

$$\frac{\partial (2Qx + d)}{\partial x} = \begin{bmatrix} 2Q_{11} + 2Q_{12} \\ 2Q_{21} + 2Q_{22} \end{bmatrix} = 2Q$$

4. test: $\hat{y}_{k+1} = X'_{k+1}(p+1) \Rightarrow \hat{y} = X'\beta$
 train: $y_{n+1} = X_{n+1}(p+1) \Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$

$$\Rightarrow \hat{y} = X'(X^T X)^{-1} X^T y \quad \text{Let } W^T = X'(X^T X)^{-1} X^T \quad \hat{y} = W^T y = \sum_{i=1}^n u_i y_i$$

KNN: $\hat{y} = \sum_{i=1}^n W_i(x) y_i \quad W_i(x) = \begin{cases} \frac{1}{k} & x_i \text{ is one of } x \text{ k nearest neighbors} \\ 0 & \text{others} \end{cases}$

Therefore, $\hat{y}$ is a special case of KNN. KNN is a form of local weight averaging. When the neighborhood expands to all dataset, it becomes ordinary linear regression.

5. We have $y \in \mathbb{R}^n \quad \hat{y} = X(X^T X)^{-1} X^T y$

We want to get $\hat{y} = X\beta$

$$\Rightarrow \text{let } \hat{\beta} = (X^T X)^{-1} X^T y$$

$$\because X = [1 \ x_0 \ x_1 \ \dots] \in \mathbb{R}^{n \times (p+1)}$$

$$\therefore \hat{\beta} \in \mathbb{R}^{p+1}$$

$$\Rightarrow \hat{y} = X\hat{\beta} \quad \hat{y} \in \text{Col}(X)$$

6. $\text{RSS} = (y_i - \hat{y}_i)^2 \Rightarrow \text{RSS}(\beta) = \|y - X\beta\|^2 = (y - X\beta)^T (y - X\beta)$

$e = y - \hat{y}$ We want to show $e \perp \text{Col}(X) \Rightarrow X^T e = 0$

$$\nabla_{\beta} \text{RSS}(\beta) = -2X^T (y - X\beta) = 0$$

$$\Rightarrow X^T (y - X\beta) = 0$$

$$\Rightarrow X^T (y - \hat{y}) = 0$$

$$X^T e = 0$$

Therefore, e is orthogonal to every col of X